



Nano research projects are underway to make use of electrodes made from nanowires to enable flat panel displays

volatile memories—molecular structures, metallic nanodots, organic molecules, nanostructured materials and so on. Although their applications are not yet mature, it is still an interesting avenue of research.”

In making existing technologies better. “The developments in nanotechnology as a sector have already shown us that this new technology is not only about doing things better, faster at a smaller scale, but also about adding new functionality to existing technologies,” says Kolstov.

“The devices, sensors, etc, are gaining another option, like, for example, the development of Spin-FET. In that case the charge and spin of the electrons are used to offer novel functionality. I think the advent of CNT electronics applications, Spin-FETs and properties of graphene are the hottest topics in industry at the moment. I am sure this may change, but in the meantime there is a lot to research in those areas,” he adds.

For healthcare and environment. Nano today is about size and material science innovations. In the future it will be about designing matter at the atomic level to address climate change, hunger, war, healthcare and energy needs,” believes Dr Canton. In the future, convergence of nanoelectronics with bioelectronics could be important for health and comfort applications, provided the technology becomes affordable. “Already, nano wires are

being used to restore movement in crippled legs, by restoring neural pathways to connect the brain to the body for movement,” informs Dr Canton. A few more possibilities include nano-energy development or storage, nanogeoeengineering to clean up the planet, nano-machines to enhance the food supply to resolve world hunger, nano-intelligence to enhance humans and to use nano-devices to deliver drugs or medicines to help heal people.

Faster data transfer between devices and networks. According to nanoforum.org, optoelectronics can help in dramatically increasing data transfer rates within devices like PCs by replacing the existing copper wiring. Instead, in the future, quantum dot-based lasers may also be used to transfer information between components within devices at the speed of light, with each piece of information ‘coded’ using a unique wavelength of light.

If we look at external networks, data transfer can take place more rapidly between two points if we increase the number of nodes in information networks. This will become possible through the development of cheap ambient-sensor networks based on nanotechnology, and will help the telecommunication sector to achieve better data transfer rates.

Circuits for wireless devices. IBM Research scientists recently announced the first integrated circuit fabricated from wafer-size graphene, and demonstrated a broadband frequency-mixer operating at frequencies up to 10GHz (10 billion cycles/second). Designed for wireless communications, this graphene-based analogue integrated circuit (having the thickness of an atom) could improve today’s wireless devices and points to the potential for a new set of applications.

At today’s conventional frequen-

cies, cellphone and transceiver signals could be improved, potentially allowing phones to work at places where they can’t today. While, at much higher frequencies, military and medical personnel could see concealed weapons or conduct medical imaging without the radiation dangers of X-rays.

“Nano is fundamentally a convergent science. Nano-Neuro-Bio-IT-Quantum all together as the top convergent sciences will transform the future of our world and our planet for the better,” foresees Dr Canton. Going forward, the nanotech enterprises will provide the ultimate convergence of computers, networks and biotech, and create products never before even imagined.

Other than the present uses for coatings, materials or components to achieve a smaller size, he foresees a few more areas where the technology may eventually have useful application. He says, “Future applications are for self-assembly of energy-on-demand for devices, robot bodies to morph to adapt to the environment or the job. Nano memory metal has attributes of body morphing. Time/space transformation, the development of neuro-brains to augment device intelligence or to augment human and machine intelligence are a few other interesting application areas.”

Going forward

While it won’t be wrong to say that we are still far from unleashing and leveraging the true potential of nanotechnology, the progress made so far in the science research labs and industrial R&D units is encouraging. These developments point to the fact that nanotechnology is truly going to be the key technology that will usher in an industrial revolution of the 21st century. Quoting the words of Michiharu Nakamura, executive VP at Hitachi, “... those who control nanotechnology will lead the industry.” ●

Vandana Sharma is executive editor for EPY’s *Linux For You* magazine, while Janani Gopalakrishnan Vikram is a technically-qualified freelance writer, editor and hands-on mom based in Singapore

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